Exercise for the Lecture
NLP with Deep Learning

TrustHLT

Dr. Prof. Ivan Habernal, Yassine Thlija
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Exercise Sheet 1

Exercise categories

The exercises are divided into two categories:

[Essential] These exercises cover concepts and skills that you are expected to understand and master.

[Training] These exercises provide more challenging problems for deepening and applying the essential concepts.

Classification Evaluation

Accuracy, precision, recall, and F1 are common measures for evaluating classification systems. In NLP, choosing an appropriate evaluation measure is particularly important because class distributions are often imbalanced.

Assume we build a simple Part-of-Speech classifier operating on the three tags `NN`, `VB`, and `ADJ`.

Exercise 1.1: Reading a Confusion Matrix

[Essential]

Consider the following confusion matrix:

		Predicted class		
		`NN`	`VB`	`ADJ`
True class	`NN`	25	5	1
	`VB`	2	15	12
	`ADJ`	1	6	0

For each class:

1. Identify TP, FP, and FN.
2. Compute precision.
3. Compute recall.
4. Compute F1.

Also compute the overall accuracy.

For class i:

$$P_i = \frac{\text{TP}}{\text{TP} + \text{FP}},$$
$$R_i = \frac{\text{TP}}{\text{TP} + \text{FN}},$$
$$F1_i = \frac{2P_iR_i}{P_i + R_i}.$$

What should happen to F1 when both precision and recall are zero?

Exercise 1.2: Macro and Micro Averaging

[Essential]

Using the confusion matrix from Exercise 1.1:

1. Compute macro precision.
2. Compute macro recall.
3. Compute macro-F1 by averaging the three per-class F1 scores.
4. Explain in words what macro averaging means.
5. What is micro-F1 for this single-label multiclass classification problem? How does it relate to accuracy?

Exercise 1.3: Choosing an Evaluation Metric

[Essential]

For each scenario below, discuss whether accuracy, macro-F1, or both should be reported. Briefly justify your answer.

1. A sentiment dataset containing approximately the same number of positive and negative examples.
2. A toxicity classifier where only 2% of examples are toxic.
3. A POS tagger where common tags occur millions of times but some linguistically interesting tags are rare.

Exercise 1.4: The Accuracy Trap

[Training]

A test dataset contains 990 examples of `classA` and 10 examples of `classB`.

Two classifiers are evaluated:

- `Model-One` predicts `classA` for every example.
- `Model-Two` predicts each class independently with probability 0.5.

For both systems:

1. Write down the expected confusion matrix.
2. Compute accuracy.
3. Compute precision, recall, and F1 for both classes.
4. Compute macro-F1.
5. Compare the systems. Which metric tells the most useful story about their behavior?

Exercise 1.5: Implementing Evaluation

[Training]

Implement a confusion matrix from scratch in Python. You may use `numpy`, but do not use an evaluation library.

Your function should accept

```
y_true = [...]
y_pred = [...]
labels = [...]
```

and return a confusion matrix.

Then implement:

1. per-class precision,
2. per-class recall,
3. per-class F1,
4. macro-F1,
5. micro-F1.

Consider explicitly what your implementation should do when a denominator is zero.

Exercise 1.6: Comparing Against scikit-learn

[Training]

Use the corresponding metrics from `scikit-learn` to verify your implementation:

`https://scikit-learn.org/stable/modules/model_evaluation.html`

Experiment with the following values of the `average` argument:

- `micro`,
- `macro`,
- `weighted`.

Construct a small imbalanced dataset for which macro-F1 and weighted-F1 differ substantially.

Explain why they differ.

Text Generation Evaluation

Unlike classification, text generation often has no single correct output. A generated sentence may be perfectly valid even if it differs from a reference sentence. This makes automatic evaluation considerably harder.

Exercise 1.7: Reference-Based Evaluation

[Essential]

Consider the reference:

> *The cat is sitting on the mat.*

and two system outputs:

> **A:** *The cat is sitting on the mat.*
>
> **B:** *A cat sits on the mat.*

Without calculating exact scores:

1. Which output should obtain the higher BLEU score?
2. Which should obtain the higher ROUGE score?
3. Is output B necessarily worse than output A?
4. What does this example tell us about reference-based evaluation?

Exercise 1.8: BLEU versus ROUGE

[Essential]

Answer briefly:

1. What does BLEU primarily measure?
2. What does ROUGE primarily measure?
3. Why was BLEU historically popular for machine translation?
4. Why was ROUGE historically popular for summarization?
5. Can either metric directly determine whether a generated statement is factually correct?

Exercise 1.9: When Overlap Metrics Fail

[Training]

Consider the reference:

> *Berlin is the capital of Germany.*

and the following outputs:

> **A:** *Berlin is the capital of Germany.*
>
> **B:** *Germany's capital is Berlin.*
>
> **C:** *Berlin is not the capital of Germany.*

Discuss:

1. Which outputs have high lexical overlap with the reference?
2. Which outputs are semantically correct?
3. Why might an overlap-based metric give an undesirable ranking?
4. What property of language makes generation evaluation harder than ordinary classification evaluation?

Exercise 1.10: Metric Design

[Training]

Suppose you are evaluating a system that generates answers to factual questions.

Would BLEU or ROUGE alone be sufficient?

Propose at least three properties that an ideal evaluation method should consider.

For each property, explain why lexical overlap with a reference may fail to measure it.

LLMs as Judges

Modern NLP systems are increasingly evaluated using another language model as an evaluator, commonly called an *LLM-as-a-Judge*.

Instead of comparing outputs only through lexical overlap, a judge model receives one or more generated answers and evaluates them according to specified criteria.

Exercise 1.11: What Is an LLM-as-a-Judge?

[Essential]

Suppose we ask:

> *Why do leaves appear green?*

A system generates:

> *Leaves appear green because chlorophyll absorbs much of the red and blue light while reflecting more green light.*

Instead of computing BLEU or ROUGE, we give the question and answer to a powerful language model and ask it to assign a score from 1 to 5 for correctness.

Answer the following:

1. What information can the LLM judge use that BLEU cannot?

2. What is one major advantage of this evaluation method?
3. What is one major risk?
4. Does using a stronger LLM automatically make the evaluation objective? Explain.

Exercise 1.12: Pointwise and Pairwise Evaluation

[Essential]

Consider two evaluation procedures.

Procedure A

> Rate the following answer from 1 to 5 for factual correctness.

Procedure B

> Given answers A and B, determine which answer is more factually correct. Respond with A, B, or Tie.

1. Which procedure is pointwise evaluation?
2. Which procedure is pairwise evaluation?
3. Give one advantage of pairwise evaluation.
4. Give one disadvantage of pairwise evaluation.

Exercise 1.13: Designing a Judge Prompt

[Essential]

Suppose an LLM should judge generated summaries.

Which of the following prompts is preferable?

Prompt A

> Is this summary good? Give it a score.

Prompt B

> Evaluate the summary on factual consistency, coverage of the important information, and conciseness. Score each criterion from 1 to 5. Base your evaluation only on the provided source document.

Explain your choice.

What additional information or instructions could make the evaluation more reproducible?

Exercise 1.14: Position Bias

[Training]

An LLM judge compares two answers, A and B.

When prompted with

> **Answer A:** X
>
> **Answer B:** Y

the judge chooses A.

After swapping the order,

> **Answer A:** Y
>
> **Answer B:** X

the judge again chooses A.

1. What problem does this experiment reveal?
2. How could you modify the evaluation procedure to detect this systematically?
3. How could you aggregate judgments from both orders?
4. What should happen if the judge gives contradictory decisions?

Exercise 1.15: Judge Biases

[Training]

For each situation, identify a potential problem with LLM-based evaluation.

1. Two answers are equally correct, but one is three times longer and receives a higher score.
2. A model is used to judge outputs generated by the same model family and consistently prefers them.
3. A stylistically polished answer containing a subtle factual error beats a terse but correct answer.
4. Changing "Rate the answer" to "Be a strict evaluator and rate the answer" substantially changes the scores.

For each problem, suggest one experiment that could reveal or quantify the effect.

Exercise 1.16: Can We Trust the Judge?

[Training]

You evaluate 500 generated answers using an LLM judge.

To test the judge, you also ask human annotators to evaluate a subset of 100 answers. The LLM agrees with the human majority on 78 of them.

Discuss the following:

1. Is 78% agreement enough to conclude that the judge is reliable?
2. What additional information would you want before trusting the judge?
3. Why should human–human agreement also be measured?
4. Would high agreement guarantee that the judge is unbiased?

Exercise 1.17: Design an Evaluation Pipeline [Training]

You have three LLMs that answer open-ended factual questions:

$$M_A, \qquad M_B, \qquad M_C.$$

You have 1,000 questions but no unique reference answer for each question.

Design an LLM-as-a-Judge evaluation protocol.

Your design should specify:

1. what information is shown to the judge,
2. whether you use pointwise or pairwise evaluation,
3. the evaluation criteria,
4. how you reduce position or ordering effects,
5. whether you use one judge or multiple judges,
6. how you validate the judge against humans,
7. how you aggregate the final results.

There is no single correct solution. Your goal is to design an evaluation whose assumptions, limitations, and potential biases are explicit.